TD-123 Verification Report - Group chat & notifications (Customer, Customer sub-user, Tech Domain, TD team)

Environment: https://dev.techdomain.com.au (API api.dev.techdomain.com.au) · Date: 2026-10-09 · Tool: Playwright (API + browser) · Spec: tests/td-123-group-chat-notifications.spec.js · Covers TD-123 (parent), TD-124, TD-125

1. Overall result

- **Scenarios (who creates the work order):** 238 pass / 16 fail (254 checks) - all 16 failed checks are the data-leak checks (4 per scenario); everything else passed.
- **Status walk (Accepted to Approved on WO 962):** 248 pass / 30 fail (278 checks) - every failed check is the same leak (non-team user 641 had already joined the room via the URL); chat, delivery and notifications passed at every status.
- **TD-124 (Tech Domain notified on WO submission): PASS** - owner and team member get "New Work Order Received".
- **TD-125 (customer notified on accept): PASS** - owner is notified; the sub-user is notified only when the sub-user created the ticket, which is the agreed rule.
- **TD-123 (group chat: owner + TD owner + TD team, plus the sub-user only when the sub-user created the work order): PASS** for chat membership, delivery and notifications in all four scenarios.
- **Customer notification when Tech Domain creates (publishes) the work order: PASS** in the final run - owner gets "Work Order Published" (S3, S4). Note: in the earlier run at 15:33 the same step produced no notification; it worked from the 16:25 run onwards, so a fix was probably deployed in between. Re-check once more before closing.
- **Data leak: FAIL** - any Tech Domain company user (not in the WO team) can join and read the customer conversation by opening its URL and then receives its notifications.

2. Test accounts used

Role	User id	Email	Password
Customer owner (Customer Company 10090955, customer of Tech Domain)	638	customer-10090955@mailinator.com	B4!Z7z9pPkuz
Customer sub-user	639	td123-subuser-1009101529@mailinator.com	Td123Pass@2026x
Tech Domain owner	27	istkhar+2@techdomain.com.au	Hussain@512
Tech Domain team member (Service Delivery 3)	640	td123-tdteammember-1009101529@mailinator.com	Td123Pass@2026x
Tech Domain user in NO team (leak control)	641	td123-tdnonteamuser-1009101730@mailinator.com	Td123Pass@2026x
Other-company customer, Neomeric (leak control)	617	td36-customer-0929@mailinator.com	Td36Pass@123
Vendor owner (leak control)	609	td-demo-company-0925@mailinator.com	TdDemo@2026x
Individual engineer (leak control; assigned engineer in the status walk)	575	Ha@mailinator.com	Hacktime123@

Other members of teams Service Delivery 3 / Operation 3 (no credentials, checked only as room participants): 263, 473, 476.

3. What was tested

- Four scenarios, one per creator: S1 Customer owner, S2 Customer sub-user, S3 Tech Domain owner, S4 Tech Domain team member. Each creates a work order for entity 104.
- Rule under test (agreed with the tester): the chat room must contain the customer owner, the TD owner and the team members (263, 473, 476, 640). The customer sub-user is included ONLY when the sub-user created the work order; otherwise the sub-user must get no chat, no messages and no notifications (checked like an outsider).
- "Work order created" notification: who received it (creator must not get their own).
- Message delivery: owner, sub-user, TD owner and team member each send one message (REST inbox send = same effect as the web socket send). Every participant must see all messages and have a MESSAGE notification for the other senders' messages, none for their own.
- Leak checks on 4 outsiders (TD non-team user, Neomeric customer, vendor, engineer): room not in inbox, conversation and room API forbidden, cannot send, no MESSAGE notification, no WO notification.
- Accept (S1, S2): Tech Domain approves the customer ticket; the owner must be notified, and the sub-user only when the sub-user created the ticket (TD-125).
- Browser evidence: sub-user, team member and non-team user log in, open the chat page and the bell.
- Status walk on WO 962: Accepted -> Requested -> Counter -> Assigned -> Confirmed -> Check-in -> Check-out -> Work done -> Verified -> Approved. At every status: room participants, 4-way message round-trip with notifications, outsider access, and who

received the status notification.

4. Scenario results (every check)

S1 - Customer owner creates the work order · WO 965 (WO-10959) · room 1748 · 64 pass / 4 fail (68 checks)

Room participants: actual [27,263,473,476,638,640] · expected [27,263,473,476,638,640] · missing [] · extra []
"Work order created" notification received by:

- **TD owner**: New Work Order Received - WO-10959 - Review the work order to accept or respond to the request.
- **TD team member**: New Work Order Received - WO-10959 - Review the work order to accept or respond to the request.

Accept notification received by:

- **Customer owner**: Work Order Accepted – WO-10959 - Your work order has been accepted and is now being processed.

- **PASS** work order created
- **PASS** customer <-> Tech Domain room exists
- **PASS** room participants = customer owner + TD owner + team (sub-user NOT included)
- **PASS** customer sub-user is NOT a participant (did not create the work order)
- **PASS** non-team company user is NOT a participant
- **PASS** Customer owner can open the room
- **PASS** TD owner can open the room
- **PASS** TD team member can open the room
- **PASS** creator did not get a notification for their own action
- **PASS** Tech Domain owner notified of the new ticket (TD-124)
- **PASS** TD non-team user got NO notification for this work order
- **PASS** Customer sub-user (not creator) got NO notification for this work order
- **PASS** Other-company customer (Neomeric) got NO notification for this work order
- **PASS** Vendor owner got NO notification for this work order
- **PASS** Individual engineer got NO notification for this work order
- **PASS** Customer owner can send a message
- **PASS** TD owner can send a message
- **PASS** TD team member can send a message
- **PASS** Customer owner sees every sent message
- **PASS** Customer owner has a MESSAGE notification for each other sender's message
- **PASS** Customer owner has no notification for their own message
- **PASS** TD owner sees every sent message
- **PASS** TD owner has a MESSAGE notification for each other sender's message
- **PASS** TD owner has no notification for their own message
- **PASS** TD team member sees every sent message
- **PASS** TD team member has a MESSAGE notification for each other sender's message
- **PASS** TD team member has no notification for their own message
- **PASS** TD non-team user: room not in inbox
- **PASS** TD non-team user: conversation forbidden
- **PASS** TD non-team user: room messages forbidden
- **PASS** TD non-team user: cannot send
- **PASS** TD non-team user: no MESSAGE notification for this room
- **PASS** Customer sub-user (not creator): room not in inbox
- **PASS** Customer sub-user (not creator): conversation forbidden
- **PASS** Customer sub-user (not creator): room messages forbidden
- **PASS** Customer sub-user (not creator): cannot send
- **PASS** Customer sub-user (not creator): no MESSAGE notification for this room
- **PASS** Other-company customer (Neomeric): room not in inbox
- **PASS** Other-company customer (Neomeric): conversation forbidden
- **PASS** Other-company customer (Neomeric): room messages forbidden
- **PASS** Other-company customer (Neomeric): cannot send
- **PASS** Other-company customer (Neomeric): no MESSAGE notification for this room
- **PASS** Vendor owner: room not in inbox
- **PASS** Vendor owner: conversation forbidden
- **PASS** Vendor owner: room messages forbidden
- **PASS** Vendor owner: cannot send
- **PASS** Vendor owner: no MESSAGE notification for this room
- **PASS** Individual engineer: room not in inbox
- **PASS** Individual engineer: conversation forbidden

- **PASS** Individual engineer: room messages forbidden
- **PASS** Individual engineer: cannot send
- **PASS** Individual engineer: no MESSAGE notification for this room
- **PASS** no LEAK TEST message landed in the room
- **PASS** work order accepted by Tech Domain
- **PASS** customer owner notified of acceptance (TD-125)
- **PASS** customer sub-user (not creator) NOT notified of acceptance
- **PASS** TD non-team user: no acceptance notification (marketplace broadcast ignored)
- **PASS** Other-company customer (Neomeric): no acceptance notification (marketplace broadcast ignored)
- **PASS** Vendor owner: no acceptance notification (marketplace broadcast ignored)
- **PASS** Individual engineer: no acceptance notification (marketplace broadcast ignored)
- **PASS** UI: sub-user (not creator) does not get the chat
- **PASS** UI: TD team member sees every sent message in the chat page
- **FAIL** UI: non-team user does not get the customer conversation
- **PASS** after opening the page the sub-user (not creator) is still NOT a participant
- **PASS** sub-user (not creator) got NO MESSAGE notification for the probe message
- **FAIL** after opening the chat URL the non-team user is still NOT a participant
- **FAIL** after opening the chat URL the non-team user still cannot read the conversation
- **FAIL** after opening the chat URL the non-team user gets NO notification for new messages

S2 - Customer sub-user creates the work order · WO 966 (WO-10960) · room 1749 · 60 pass / 4 fail (64 checks)

Room participants: actual [27,263,473,476,638,639,640] · expected [27,263,473,476,638,639,640] · missing [] · extra []
"Work order created" notification received by:

- **TD owner**: New Work Order Received - WO-10960 - Review the work order to accept or respond to the request.
- **TD team member**: New Work Order Received - WO-10960 - Review the work order to accept or respond to the request.

Accept notification received by:

- **Customer owner**: Work Order Accepted – WO-10960 - Your work order has been accepted and is now being processed.
- **Customer sub-user**: Work Order Accepted – WO-10960 - Your work order has been accepted and is now being processed.

- **PASS** work order created
- **PASS** customer <-> Tech Domain room exists
- **PASS** room participants = customer owner + sub-user (creator) + TD owner + team
- **PASS** non-team company user is NOT a participant
- **PASS** Customer owner can open the room
- **PASS** Customer sub-user can open the room
- **PASS** TD owner can open the room
- **PASS** TD team member can open the room
- **PASS** creator did not get a notification for their own action
- **PASS** Tech Domain owner notified of the new ticket (TD-124)
- **PASS** TD non-team user got NO notification for this work order
- **PASS** Other-company customer (Neomeric) got NO notification for this work order
- **PASS** Vendor owner got NO notification for this work order
- **PASS** Individual engineer got NO notification for this work order
- **PASS** Customer owner can send a message
- **PASS** Customer sub-user can send a message
- **PASS** TD owner can send a message
- **PASS** TD team member can send a message
- **PASS** Customer owner sees every sent message
- **PASS** Customer owner has a MESSAGE notification for each other sender's message
- **PASS** Customer owner has no notification for their own message
- **PASS** Customer sub-user sees every sent message
- **PASS** Customer sub-user has a MESSAGE notification for each other sender's message
- **PASS** Customer sub-user has no notification for their own message
- **PASS** TD owner sees every sent message
- **PASS** TD owner has a MESSAGE notification for each other sender's message
- **PASS** TD owner has no notification for their own message
- **PASS** TD team member sees every sent message
- **PASS** TD team member has a MESSAGE notification for each other sender's message
- **PASS** TD team member has no notification for their own message
- **PASS** TD non-team user: room not in inbox
- **PASS** TD non-team user: conversation forbidden

- **PASS** TD non-team user: room messages forbidden
- **PASS** TD non-team user: cannot send
- **PASS** TD non-team user: no MESSAGE notification for this room
- **PASS** Other-company customer (Neomeric): room not in inbox
- **PASS** Other-company customer (Neomeric): conversation forbidden
- **PASS** Other-company customer (Neomeric): room messages forbidden
- **PASS** Other-company customer (Neomeric): cannot send
- **PASS** Other-company customer (Neomeric): no MESSAGE notification for this room
- **PASS** Vendor owner: room not in inbox
- **PASS** Vendor owner: conversation forbidden
- **PASS** Vendor owner: room messages forbidden
- **PASS** Vendor owner: cannot send
- **PASS** Vendor owner: no MESSAGE notification for this room
- **PASS** Individual engineer: room not in inbox
- **PASS** Individual engineer: conversation forbidden
- **PASS** Individual engineer: room messages forbidden
- **PASS** Individual engineer: cannot send
- **PASS** Individual engineer: no MESSAGE notification for this room
- **PASS** no LEAK TEST message landed in the room
- **PASS** work order accepted by Tech Domain
- **PASS** customer owner notified of acceptance (TD-125)
- **PASS** customer sub-user (creator) notified of acceptance (TD-125)
- **PASS** TD non-team user: no acceptance notification (marketplace broadcast ignored)
- **PASS** Other-company customer (Neomeric): no acceptance notification (marketplace broadcast ignored)
- **PASS** Vendor owner: no acceptance notification (marketplace broadcast ignored)
- **PASS** Individual engineer: no acceptance notification (marketplace broadcast ignored)
- **PASS** UI: Customer sub-user sees every sent message in the chat page
- **PASS** UI: TD team member sees every sent message in the chat page
- **FAIL** UI: non-team user does not get the customer conversation
- **FAIL** after opening the chat URL the non-team user is still NOT a participant
- **FAIL** after opening the chat URL the non-team user still cannot read the conversation
- **FAIL** after opening the chat URL the non-team user gets NO notification for new messages

S3 - Tech Domain owner creates the work order · WO 967 (WO-10961) · room 1750 · 57 pass / 4 fail (61 checks)

Room participants: actual [27,263,473,476,638,640] · expected [27,263,473,476,638,640] · missing [] · extra []
"Work order created" notification received by:

- **Customer owner**: Work Order Published – WO-10961 - Your work order has been published and is now being processed

- **PASS** work order created
- **PASS** customer <-> Tech Domain room exists
- **PASS** room participants = customer owner + TD owner + team (sub-user NOT included)
- **PASS** customer sub-user is NOT a participant (did not create the work order)
- **PASS** non-team company user is NOT a participant
- **PASS** Customer owner can open the room
- **PASS** TD owner can open the room
- **PASS** TD team member can open the room
- **PASS** creator did not get a notification for their own action
- **PASS** customer owner notified of the new work order
- **PASS** TD non-team user got NO notification for this work order
- **PASS** Customer sub-user (not creator) got NO notification for this work order
- **PASS** Other-company customer (Neomeric) got NO notification for this work order
- **PASS** Vendor owner got NO notification for this work order
- **PASS** Individual engineer got NO notification for this work order
- **PASS** Customer owner can send a message
- **PASS** TD owner can send a message
- **PASS** TD team member can send a message
- **PASS** Customer owner sees every sent message
- **PASS** Customer owner has a MESSAGE notification for each other sender's message
- **PASS** Customer owner has no notification for their own message
- **PASS** TD owner sees every sent message

- **PASS** TD owner has a MESSAGE notification for each other sender's message
- **PASS** TD owner has no notification for their own message
- **PASS** TD team member sees every sent message
- **PASS** TD team member has a MESSAGE notification for each other sender's message
- **PASS** TD team member has no notification for their own message
- **PASS** TD non-team user: room not in inbox
- **PASS** TD non-team user: conversation forbidden
- **PASS** TD non-team user: room messages forbidden
- **PASS** TD non-team user: cannot send
- **PASS** TD non-team user: no MESSAGE notification for this room
- **PASS** Customer sub-user (not creator): room not in inbox
- **PASS** Customer sub-user (not creator): conversation forbidden
- **PASS** Customer sub-user (not creator): room messages forbidden
- **PASS** Customer sub-user (not creator): cannot send
- **PASS** Customer sub-user (not creator): no MESSAGE notification for this room
- **PASS** Other-company customer (Neomeric): room not in inbox
- **PASS** Other-company customer (Neomeric): conversation forbidden
- **PASS** Other-company customer (Neomeric): room messages forbidden
- **PASS** Other-company customer (Neomeric): cannot send
- **PASS** Other-company customer (Neomeric): no MESSAGE notification for this room
- **PASS** Vendor owner: room not in inbox
- **PASS** Vendor owner: conversation forbidden
- **PASS** Vendor owner: room messages forbidden
- **PASS** Vendor owner: cannot send
- **PASS** Vendor owner: no MESSAGE notification for this room
- **PASS** Individual engineer: room not in inbox
- **PASS** Individual engineer: conversation forbidden
- **PASS** Individual engineer: room messages forbidden
- **PASS** Individual engineer: cannot send
- **PASS** Individual engineer: no MESSAGE notification for this room
- **PASS** no LEAK TEST message landed in the room
- **PASS** UI: sub-user (not creator) does not get the chat
- **PASS** UI: TD team member sees every sent message in the chat page
- **FAIL** UI: non-team user does not get the customer conversation
- **PASS** after opening the page the sub-user (not creator) is still NOT a participant
- **PASS** sub-user (not creator) got NO MESSAGE notification for the probe message
- **FAIL** after opening the chat URL the non-team user is still NOT a participant
- **FAIL** after opening the chat URL the non-team user still cannot read the conversation
- **FAIL** after opening the chat URL the non-team user gets NO notification for new messages

S4 - Tech Domain team member creates the work order · WO 968 (WO-10962) · room 1751 · 57 pass / 4 fail (61 checks)

Room participants: actual [27,263,473,476,638,640] · expected [27,263,473,476,638,640] · missing [] · extra []
"Work order created" notification received by:

- **Customer owner**: Work Order Published – WO-10962 - Your work order has been published and is now being processed

- **PASS** work order created
- **PASS** customer <-> Tech Domain room exists
- **PASS** room participants = customer owner + TD owner + team (sub-user NOT included)
- **PASS** customer sub-user is NOT a participant (did not create the work order)
- **PASS** non-team company user is NOT a participant
- **PASS** Customer owner can open the room
- **PASS** TD owner can open the room
- **PASS** TD team member can open the room
- **PASS** creator did not get a notification for their own action
- **PASS** customer owner notified of the new work order
- **PASS** TD non-team user got NO notification for this work order
- **PASS** Customer sub-user (not creator) got NO notification for this work order
- **PASS** Other-company customer (Neomeric) got NO notification for this work order
- **PASS** Vendor owner got NO notification for this work order
- **PASS** Individual engineer got NO notification for this work order

- **PASS** Customer owner can send a message
- **PASS** TD owner can send a message
- **PASS** TD team member can send a message
- **PASS** Customer owner sees every sent message
- **PASS** Customer owner has a MESSAGE notification for each other sender's message
- **PASS** Customer owner has no notification for their own message
- **PASS** TD owner sees every sent message
- **PASS** TD owner has a MESSAGE notification for each other sender's message
- **PASS** TD owner has no notification for their own message
- **PASS** TD team member sees every sent message
- **PASS** TD team member has a MESSAGE notification for each other sender's message
- **PASS** TD team member has no notification for their own message
- **PASS** TD non-team user: room not in inbox
- **PASS** TD non-team user: conversation forbidden
- **PASS** TD non-team user: room messages forbidden
- **PASS** TD non-team user: cannot send
- **PASS** TD non-team user: no MESSAGE notification for this room
- **PASS** Customer sub-user (not creator): room not in inbox
- **PASS** Customer sub-user (not creator): conversation forbidden
- **PASS** Customer sub-user (not creator): room messages forbidden
- **PASS** Customer sub-user (not creator): cannot send
- **PASS** Customer sub-user (not creator): no MESSAGE notification for this room
- **PASS** Other-company customer (Neomeric): room not in inbox
- **PASS** Other-company customer (Neomeric): conversation forbidden
- **PASS** Other-company customer (Neomeric): room messages forbidden
- **PASS** Other-company customer (Neomeric): cannot send
- **PASS** Other-company customer (Neomeric): no MESSAGE notification for this room
- **PASS** Vendor owner: room not in inbox
- **PASS** Vendor owner: conversation forbidden
- **PASS** Vendor owner: room messages forbidden
- **PASS** Vendor owner: cannot send
- **PASS** Vendor owner: no MESSAGE notification for this room
- **PASS** Individual engineer: room not in inbox
- **PASS** Individual engineer: conversation forbidden
- **PASS** Individual engineer: room messages forbidden
- **PASS** Individual engineer: cannot send
- **PASS** Individual engineer: no MESSAGE notification for this room
- **PASS** no LEAK TEST message landed in the room
- **PASS** UI: sub-user (not creator) does not get the chat
- **PASS** UI: TD team member sees every sent message in the chat page
- **FAIL** UI: non-team user does not get the customer conversation
- **PASS** after opening the page the sub-user (not creator) is still NOT a participant
- **PASS** sub-user (not creator) got NO MESSAGE notification for the probe message
- **FAIL** after opening the chat URL the non-team user is still NOT a participant
- **FAIL** after opening the chat URL the non-team user still cannot read the conversation
- **FAIL** after opening the chat URL the non-team user gets NO notification for new messages

5. Status walk results (WO 962 / WO-10956, created by the customer sub-user)

- Reached: {"assigned":true,"confirmed":true,"checkIn":true,"checkOut":true,"complete":true,"verified":true,"approved":true}
- Route taken: engineer request (REQUESTED) -> engineer counter offer (COUNTER) -> client accepts counter (= ASSIGNED) -> client sets CONFIRMED -> schedule moved to now -> engineer check-in / check-out -> MARK_COMPLETE (WORKDONE) -> client REVIEWED + verify (VERIFIED) -> client approve (APPROVED).

ACCEPTED (published) · WO status PUBLISHED / - · 24 pass / 3 fail (27 checks)

Status notification received by:
- nobody

- **PASS** work order status is PUBLISHED
- **PASS** customer room still exists
- **FAIL** customer room participants unchanged (owner + sub-user + TD owner + team)
- **PASS** engineer is NOT in the customer room
- **PASS** Customer owner can send at ACCEPTED (published)
- **PASS** TD team member can send at ACCEPTED (published)
- **PASS** Customer sub-user can send at ACCEPTED (published)
- **PASS** TD owner can send at ACCEPTED (published)
- **PASS** Customer owner sees every sent ACCEPTED (published) message
- **PASS** Customer owner notified of the other 3 ACCEPTED (published) messages
- **PASS** Customer sub-user sees every sent ACCEPTED (published) message
- **PASS** Customer sub-user notified of the other 3 ACCEPTED (published) messages
- **PASS** TD owner sees every sent ACCEPTED (published) message
- **PASS** TD owner notified of the other 3 ACCEPTED (published) messages
- **PASS** TD team member sees every sent ACCEPTED (published) message
- **PASS** TD team member notified of the other 3 ACCEPTED (published) messages
- **PASS** Individual engineer: customer room forbidden at ACCEPTED (published)
- **PASS** Individual engineer: no MESSAGE notification from the customer room at ACCEPTED (published)
- **FAIL** TD non-team user: customer room forbidden at ACCEPTED (published)
- **FAIL** TD non-team user: no MESSAGE notification from the customer room at ACCEPTED (published)
- **PASS** Other-company customer (Neomeric): customer room forbidden at ACCEPTED (published)
- **PASS** Other-company customer (Neomeric): no MESSAGE notification from the customer room at ACCEPTED (published)
- **PASS** Vendor owner: customer room forbidden at ACCEPTED (published)
- **PASS** Vendor owner: no MESSAGE notification from the customer room at ACCEPTED (published)
- **PASS** TD non-team user: no status notification at ACCEPTED (published) (marketplace broadcast ignored)
- **PASS** Other-company customer (Neomeric): no status notification at ACCEPTED (published) (marketplace broadcast ignored)
- **PASS** Vendor owner: no status notification at ACCEPTED (published) (marketplace broadcast ignored)

REQUESTED · WO status PUBLISHED / - · 24 pass / 3 fail (27 checks)

Status notification received by:
- **TD owner**: WO-10956: TD123 S2 1009102952 by Customer sub-user - Syed Hamza has sent an offer.
- **TD team member**: WO-10956: TD123 S2 1009102952 by Customer sub-user - Syed Hamza has sent an offer.

- **PASS** customer room still exists
- **FAIL** customer room participants unchanged (owner + sub-user + TD owner + team)
- **PASS** engineer is NOT in the customer room
- **PASS** Customer owner can send at REQUESTED
- **PASS** TD team member can send at REQUESTED
- **PASS** Customer sub-user can send at REQUESTED
- **PASS** TD owner can send at REQUESTED
- **PASS** Customer owner sees every sent REQUESTED message
- **PASS** Customer owner notified of the other 3 REQUESTED messages
- **PASS** Customer sub-user sees every sent REQUESTED message
- **PASS** Customer sub-user notified of the other 3 REQUESTED messages
- **PASS** TD owner sees every sent REQUESTED message
- **PASS** TD owner notified of the other 3 REQUESTED messages
- **PASS** TD team member sees every sent REQUESTED message
- **PASS** TD team member notified of the other 3 REQUESTED messages
- **PASS** Individual engineer: customer room forbidden at REQUESTED

- **PASS** Individual engineer: no MESSAGE notification from the customer room at REQUESTED
- **FAIL** TD non-team user: customer room forbidden at REQUESTED
- **FAIL** TD non-team user: no MESSAGE notification from the customer room at REQUESTED
- **PASS** Other-company customer (Neomeric): customer room forbidden at REQUESTED
- **PASS** Other-company customer (Neomeric): no MESSAGE notification from the customer room at REQUESTED
- **PASS** Vendor owner: customer room forbidden at REQUESTED
- **PASS** Vendor owner: no MESSAGE notification from the customer room at REQUESTED
- **PASS** Tech Domain team member notified at REQUESTED
- **PASS** TD non-team user: no status notification at REQUESTED (marketplace broadcast ignored)
- **PASS** Other-company customer (Neomeric): no status notification at REQUESTED (marketplace broadcast ignored)
- **PASS** Vendor owner: no status notification at REQUESTED (marketplace broadcast ignored)

COUNTER · WO status PUBLISHED / - · 24 pass / 3 fail (27 checks)

Status notification received by:

- **TD owner**: WO-10956: TD123 S2 1009102952 by Customer sub-user - Syed Hamza has updated a counter offer.
- **TD team member**: WO-10956: TD123 S2 1009102952 by Customer sub-user - Syed Hamza has updated a counter offer.

- **PASS** customer room still exists
- **FAIL** customer room participants unchanged (owner + sub-user + TD owner + team)
- **PASS** engineer is NOT in the customer room
- **PASS** Customer owner can send at COUNTER
- **PASS** TD team member can send at COUNTER
- **PASS** Customer sub-user can send at COUNTER
- **PASS** TD owner can send at COUNTER
- **PASS** Customer owner sees every sent COUNTER message
- **PASS** Customer owner notified of the other 3 COUNTER messages
- **PASS** Customer sub-user sees every sent COUNTER message
- **PASS** Customer sub-user notified of the other 3 COUNTER messages
- **PASS** TD owner sees every sent COUNTER message
- **PASS** TD owner notified of the other 3 COUNTER messages
- **PASS** TD team member sees every sent COUNTER message
- **PASS** TD team member notified of the other 3 COUNTER messages
- **PASS** Individual engineer: customer room forbidden at COUNTER
- **PASS** Individual engineer: no MESSAGE notification from the customer room at COUNTER
- **FAIL** TD non-team user: customer room forbidden at COUNTER
- **FAIL** TD non-team user: no MESSAGE notification from the customer room at COUNTER
- **PASS** Other-company customer (Neomeric): customer room forbidden at COUNTER
- **PASS** Other-company customer (Neomeric): no MESSAGE notification from the customer room at COUNTER
- **PASS** Vendor owner: customer room forbidden at COUNTER
- **PASS** Vendor owner: no MESSAGE notification from the customer room at COUNTER
- **PASS** Tech Domain team member notified at COUNTER
- **PASS** TD non-team user: no status notification at COUNTER (marketplace broadcast ignored)
- **PASS** Other-company customer (Neomeric): no status notification at COUNTER (marketplace broadcast ignored)
- **PASS** Vendor owner: no status notification at COUNTER (marketplace broadcast ignored)

ASSIGNED · WO status ASSIGNED / NOT_CONFIRMED · 26 pass / 3 fail (29 checks)

Status notification received by:

- **Customer owner**: Engineer Assigned – WO-10956 - View the work order for engineer details and service updates.
- **Customer sub-user**: Engineer Assigned – WO-10956 - View the work order for engineer details and service updates.
- **Individual engineer**: New Work Order Assigned – WO-10956 - You have been assigned to a new work order.

- **PASS** work order status is ASSIGNED
- **PASS** customer room still exists
- **FAIL** customer room participants unchanged (owner + sub-user + TD owner + team)
- **PASS** engineer is NOT in the customer room
- **PASS** Customer owner can send at ASSIGNED
- **PASS** TD team member can send at ASSIGNED
- **PASS** Customer sub-user can send at ASSIGNED
- **PASS** TD owner can send at ASSIGNED
- **PASS** Customer owner sees every sent ASSIGNED message

- **PASS** Customer owner notified of the other 3 ASSIGNED messages
- **PASS** Customer sub-user sees every sent ASSIGNED message
- **PASS** Customer sub-user notified of the other 3 ASSIGNED messages
- **PASS** TD owner sees every sent ASSIGNED message
- **PASS** TD owner notified of the other 3 ASSIGNED messages
- **PASS** TD team member sees every sent ASSIGNED message
- **PASS** TD team member notified of the other 3 ASSIGNED messages
- **PASS** Individual engineer: customer room forbidden at ASSIGNED
- **PASS** Individual engineer: no MESSAGE notification from the customer room at ASSIGNED
- **FAIL** TD non-team user: customer room forbidden at ASSIGNED
- **FAIL** TD non-team user: no MESSAGE notification from the customer room at ASSIGNED
- **PASS** Other-company customer (Neomeric): customer room forbidden at ASSIGNED
- **PASS** Other-company customer (Neomeric): no MESSAGE notification from the customer room at ASSIGNED
- **PASS** Vendor owner: customer room forbidden at ASSIGNED
- **PASS** Vendor owner: no MESSAGE notification from the customer room at ASSIGNED
- **PASS** customer owner notified at ASSIGNED
- **PASS** customer sub-user notified at ASSIGNED
- **PASS** TD non-team user: no status notification at ASSIGNED (marketplace broadcast ignored)
- **PASS** Other-company customer (Neomeric): no status notification at ASSIGNED (marketplace broadcast ignored)
- **PASS** Vendor owner: no status notification at ASSIGNED (marketplace broadcast ignored)

CONFIRMED · WO status ASSIGNED / CONFIRMED · 25 pass / 3 fail (28 checks)

Status notification received by:

- **Customer owner**: Work Order Confirmed by Engineer – WO-10956 - The assigned engineer has confirmed the work order
- **Customer sub-user**: Work Order Confirmed by Engineer – WO-10956 - The assigned engineer has confirmed the work order
- **TD team member**: Work Order Confirmed by Engineer – WO-10956 - The assigned engineer has confirmed the work order

- **PASS** customer room still exists
- **FAIL** customer room participants unchanged (owner + sub-user + TD owner + team)
- **PASS** engineer is NOT in the customer room
- **PASS** Customer owner can send at CONFIRMED
- **PASS** TD team member can send at CONFIRMED
- **PASS** Customer sub-user can send at CONFIRMED
- **PASS** TD owner can send at CONFIRMED
- **PASS** Customer owner sees every sent CONFIRMED message
- **PASS** Customer owner notified of the other 3 CONFIRMED messages
- **PASS** Customer sub-user sees every sent CONFIRMED message
- **PASS** Customer sub-user notified of the other 3 CONFIRMED messages
- **PASS** TD owner sees every sent CONFIRMED message
- **PASS** TD owner notified of the other 3 CONFIRMED messages
- **PASS** TD team member sees every sent CONFIRMED message
- **PASS** TD team member notified of the other 3 CONFIRMED messages
- **PASS** Individual engineer: customer room forbidden at CONFIRMED
- **PASS** Individual engineer: no MESSAGE notification from the customer room at CONFIRMED
- **FAIL** TD non-team user: customer room forbidden at CONFIRMED
- **FAIL** TD non-team user: no MESSAGE notification from the customer room at CONFIRMED
- **PASS** Other-company customer (Neomeric): customer room forbidden at CONFIRMED
- **PASS** Other-company customer (Neomeric): no MESSAGE notification from the customer room at CONFIRMED
- **PASS** Vendor owner: customer room forbidden at CONFIRMED
- **PASS** Vendor owner: no MESSAGE notification from the customer room at CONFIRMED
- **PASS** customer owner notified at CONFIRMED
- **PASS** customer sub-user notified at CONFIRMED
- **PASS** TD non-team user: no status notification at CONFIRMED (marketplace broadcast ignored)
- **PASS** Other-company customer (Neomeric): no status notification at CONFIRMED (marketplace broadcast ignored)
- **PASS** Vendor owner: no status notification at CONFIRMED (marketplace broadcast ignored)

CHECK_IN (in progress) · WO status INPROGRESS / CHECK_IN · 26 pass / 3 fail (29 checks)

Status notification received by:

- **Customer owner**: Engineer Checked In – WO-10956 - The assigned engineer has checked in and ready to begin the work.
- **Customer sub-user**: Engineer Checked In – WO-10956 - The assigned engineer has checked in and ready to begin the work.
- **TD team member**: Engineer Checked In – WO-10956 - The assigned engineer has checked in and ready to begin the work.

- **Individual engineer**: WO-10956: TD123 S2 1009102952 by Customer sub-user - Schedule has been updated to 09/10/2026, 10:31 am. Please confirm.; WO-10956: TD123 S2 1009102952 by Customer sub-user - Schedule has been updated to 27/10/2026, 11:00 am. Please confirm.

- **PASS** work order status is INPROGRESS
- **PASS** customer room still exists
- **FAIL** customer room participants unchanged (owner + sub-user + TD owner + team)
- **PASS** engineer is NOT in the customer room
- **PASS** Customer owner can send at CHECK_IN (in progress)
- **PASS** TD team member can send at CHECK_IN (in progress)
- **PASS** Customer sub-user can send at CHECK_IN (in progress)
- **PASS** TD owner can send at CHECK_IN (in progress)
- **PASS** Customer owner sees every sent CHECK_IN (in progress) message
- **PASS** Customer owner notified of the other 3 CHECK_IN (in progress) messages
- **PASS** Customer sub-user sees every sent CHECK_IN (in progress) message
- **PASS** Customer sub-user notified of the other 3 CHECK_IN (in progress) messages
- **PASS** TD owner sees every sent CHECK_IN (in progress) message
- **PASS** TD owner notified of the other 3 CHECK_IN (in progress) messages
- **PASS** TD team member sees every sent CHECK_IN (in progress) message
- **PASS** TD team member notified of the other 3 CHECK_IN (in progress) messages
- **PASS** Individual engineer: customer room forbidden at CHECK_IN (in progress)
- **PASS** Individual engineer: no MESSAGE notification from the customer room at CHECK_IN (in progress)
- **FAIL** TD non-team user: customer room forbidden at CHECK_IN (in progress)
- **FAIL** TD non-team user: no MESSAGE notification from the customer room at CHECK_IN (in progress)
- **PASS** Other-company customer (Neomeric): customer room forbidden at CHECK_IN (in progress)
- **PASS** Other-company customer (Neomeric): no MESSAGE notification from the customer room at CHECK_IN (in progress)
- **PASS** Vendor owner: customer room forbidden at CHECK_IN (in progress)
- **PASS** Vendor owner: no MESSAGE notification from the customer room at CHECK_IN (in progress)
- **PASS** customer owner notified at CHECK_IN (in progress)
- **PASS** customer sub-user notified at CHECK_IN (in progress)
- **PASS** TD non-team user: no status notification at CHECK_IN (in progress) (marketplace broadcast ignored)
- **PASS** Other-company customer (Neomeric): no status notification at CHECK_IN (in progress) (marketplace broadcast ignored)
- **PASS** Vendor owner: no status notification at CHECK_IN (in progress) (marketplace broadcast ignored)

CHECK_OUT · WO status INPROGRESS / CHECK_OUT · 25 pass / 3 fail (28 checks)

Status notification received by:

- **Customer owner**: Engineer Checked Out – WO-10956 - The assigned engineer has checked out after completing the onsite activity.
- **Customer sub-user**: Engineer Checked Out – WO-10956 - The assigned engineer has checked out after completing the onsite activity.
- **TD team member**: Engineer Checked Out – WO-10956 - The assigned engineer has checked out after completing the onsite activity.

- **PASS** customer room still exists
- **FAIL** customer room participants unchanged (owner + sub-user + TD owner + team)
- **PASS** engineer is NOT in the customer room
- **PASS** Customer owner can send at CHECK_OUT
- **PASS** TD team member can send at CHECK_OUT
- **PASS** Customer sub-user can send at CHECK_OUT
- **PASS** TD owner can send at CHECK_OUT
- **PASS** Customer owner sees every sent CHECK_OUT message
- **PASS** Customer owner notified of the other 3 CHECK_OUT messages
- **PASS** Customer sub-user sees every sent CHECK_OUT message
- **PASS** Customer sub-user notified of the other 3 CHECK_OUT messages
- **PASS** TD owner sees every sent CHECK_OUT message
- **PASS** TD owner notified of the other 3 CHECK_OUT messages
- **PASS** TD team member sees every sent CHECK_OUT message
- **PASS** TD team member notified of the other 3 CHECK_OUT messages
- **PASS** Individual engineer: customer room forbidden at CHECK_OUT
- **PASS** Individual engineer: no MESSAGE notification from the customer room at CHECK_OUT
- **FAIL** TD non-team user: customer room forbidden at CHECK_OUT

- **FAIL** TD non-team user: no MESSAGE notification from the customer room at CHECK_OUT
- **PASS** Other-company customer (Neomeric): customer room forbidden at CHECK_OUT
- **PASS** Other-company customer (Neomeric): no MESSAGE notification from the customer room at CHECK_OUT
- **PASS** Vendor owner: customer room forbidden at CHECK_OUT
- **PASS** Vendor owner: no MESSAGE notification from the customer room at CHECK_OUT
- **PASS** customer owner notified at CHECK_OUT
- **PASS** customer sub-user notified at CHECK_OUT
- **PASS** TD non-team user: no status notification at CHECK_OUT (marketplace broadcast ignored)
- **PASS** Other-company customer (Neomeric): no status notification at CHECK_OUT (marketplace broadcast ignored)
- **PASS** Vendor owner: no status notification at CHECK_OUT (marketplace broadcast ignored)

WORKDONE · WO status WORKDONE / - · 25 pass / 3 fail (28 checks)

Status notification received by:
- **Customer owner**: Work Order Completed – WO-10956 - The engineer has completed the required service activities.
- **Customer sub-user**: Work Order Completed – WO-10956 - The engineer has completed the required service activities.
- **TD owner**: Work Order Completed – WO-10956 - The engineer has completed the required service activities.
- **TD team member**: Work Order Completed – WO-10956 - The engineer has completed the required service activities.

- **PASS** customer room still exists
- **FAIL** customer room participants unchanged (owner + sub-user + TD owner + team)
- **PASS** engineer is NOT in the customer room
- **PASS** Customer owner can send at WORKDONE
- **PASS** TD team member can send at WORKDONE
- **PASS** Customer sub-user can send at WORKDONE
- **PASS** TD owner can send at WORKDONE
- **PASS** Customer owner sees every sent WORKDONE message
- **PASS** Customer owner notified of the other 3 WORKDONE messages
- **PASS** Customer sub-user sees every sent WORKDONE message
- **PASS** Customer sub-user notified of the other 3 WORKDONE messages
- **PASS** TD owner sees every sent WORKDONE message
- **PASS** TD owner notified of the other 3 WORKDONE messages
- **PASS** TD team member sees every sent WORKDONE message
- **PASS** TD team member notified of the other 3 WORKDONE messages
- **PASS** Individual engineer: customer room forbidden at WORKDONE
- **PASS** Individual engineer: no MESSAGE notification from the customer room at WORKDONE
- **FAIL** TD non-team user: customer room forbidden at WORKDONE
- **FAIL** TD non-team user: no MESSAGE notification from the customer room at WORKDONE
- **PASS** Other-company customer (Neomeric): customer room forbidden at WORKDONE
- **PASS** Other-company customer (Neomeric): no MESSAGE notification from the customer room at WORKDONE
- **PASS** Vendor owner: customer room forbidden at WORKDONE
- **PASS** Vendor owner: no MESSAGE notification from the customer room at WORKDONE
- **PASS** customer owner notified at WORKDONE
- **PASS** customer sub-user notified at WORKDONE
- **PASS** TD non-team user: no status notification at WORKDONE (marketplace broadcast ignored)
- **PASS** Other-company customer (Neomeric): no status notification at WORKDONE (marketplace broadcast ignored)
- **PASS** Vendor owner: no status notification at WORKDONE (marketplace broadcast ignored)

VERIFIED · WO status VERIFIED / PO_PENDING · 23 pass / 3 fail (26 checks)

Status notification received by:
- nobody

- **PASS** customer room still exists
- **FAIL** customer room participants unchanged (owner + sub-user + TD owner + team)
- **PASS** engineer is NOT in the customer room
- **PASS** Customer owner can send at VERIFIED
- **PASS** TD team member can send at VERIFIED
- **PASS** Customer sub-user can send at VERIFIED
- **PASS** TD owner can send at VERIFIED
- **PASS** Customer owner sees every sent VERIFIED message
- **PASS** Customer owner notified of the other 3 VERIFIED messages

- **PASS** Customer sub-user sees every sent VERIFIED message
- **PASS** Customer sub-user notified of the other 3 VERIFIED messages
- **PASS** TD owner sees every sent VERIFIED message
- **PASS** TD owner notified of the other 3 VERIFIED messages
- **PASS** TD team member sees every sent VERIFIED message
- **PASS** TD team member notified of the other 3 VERIFIED messages
- **PASS** Individual engineer: customer room forbidden at VERIFIED
- **PASS** Individual engineer: no MESSAGE notification from the customer room at VERIFIED
- **FAIL** TD non-team user: customer room forbidden at VERIFIED
- **FAIL** TD non-team user: no MESSAGE notification from the customer room at VERIFIED
- **PASS** Other-company customer (Neomeric): customer room forbidden at VERIFIED
- **PASS** Other-company customer (Neomeric): no MESSAGE notification from the customer room at VERIFIED
- **PASS** Vendor owner: customer room forbidden at VERIFIED
- **PASS** Vendor owner: no MESSAGE notification from the customer room at VERIFIED
- **PASS** TD non-team user: no status notification at VERIFIED (marketplace broadcast ignored)
- **PASS** Other-company customer (Neomeric): no status notification at VERIFIED (marketplace broadcast ignored)
- **PASS** Vendor owner: no status notification at VERIFIED (marketplace broadcast ignored)

APPROVED · WO status APPROVED / PO_PENDING · 26 pass / 3 fail (29 checks)

Status notification received by:

- **Customer owner**: WO-10956: TD123 S2 1009102952 by Customer sub-user - Your work order has been approved.
- **Customer sub-user**: WO-10956: TD123 S2 1009102952 by Customer sub-user - Your work order has been approved.
- **Individual engineer**: Work Order Approved – WO-10956 - Your completed work order has been approved.

- **PASS** work order status is APPROVED
- **PASS** customer room still exists
- **FAIL** customer room participants unchanged (owner + sub-user + TD owner + team)
- **PASS** engineer is NOT in the customer room
- **PASS** Customer owner can send at APPROVED
- **PASS** TD team member can send at APPROVED
- **PASS** Customer sub-user can send at APPROVED
- **PASS** TD owner can send at APPROVED
- **PASS** Customer owner sees every sent APPROVED message
- **PASS** Customer owner notified of the other 3 APPROVED messages
- **PASS** Customer sub-user sees every sent APPROVED message
- **PASS** Customer sub-user notified of the other 3 APPROVED messages
- **PASS** TD owner sees every sent APPROVED message
- **PASS** TD owner notified of the other 3 APPROVED messages
- **PASS** TD team member sees every sent APPROVED message
- **PASS** TD team member notified of the other 3 APPROVED messages
- **PASS** Individual engineer: customer room forbidden at APPROVED
- **PASS** Individual engineer: no MESSAGE notification from the customer room at APPROVED
- **FAIL** TD non-team user: customer room forbidden at APPROVED
- **FAIL** TD non-team user: no MESSAGE notification from the customer room at APPROVED
- **PASS** Other-company customer (Neomeric): customer room forbidden at APPROVED
- **PASS** Other-company customer (Neomeric): no MESSAGE notification from the customer room at APPROVED
- **PASS** Vendor owner: customer room forbidden at APPROVED
- **PASS** Vendor owner: no MESSAGE notification from the customer room at APPROVED
- **PASS** customer owner notified at APPROVED
- **PASS** customer sub-user notified at APPROVED
- **PASS** TD non-team user: no status notification at APPROVED (marketplace broadcast ignored)
- **PASS** Other-company customer (Neomeric): no status notification at APPROVED (marketplace broadcast ignored)
- **PASS** Vendor owner: no status notification at APPROVED (marketplace broadcast ignored)

6. Findings (bugs to report)

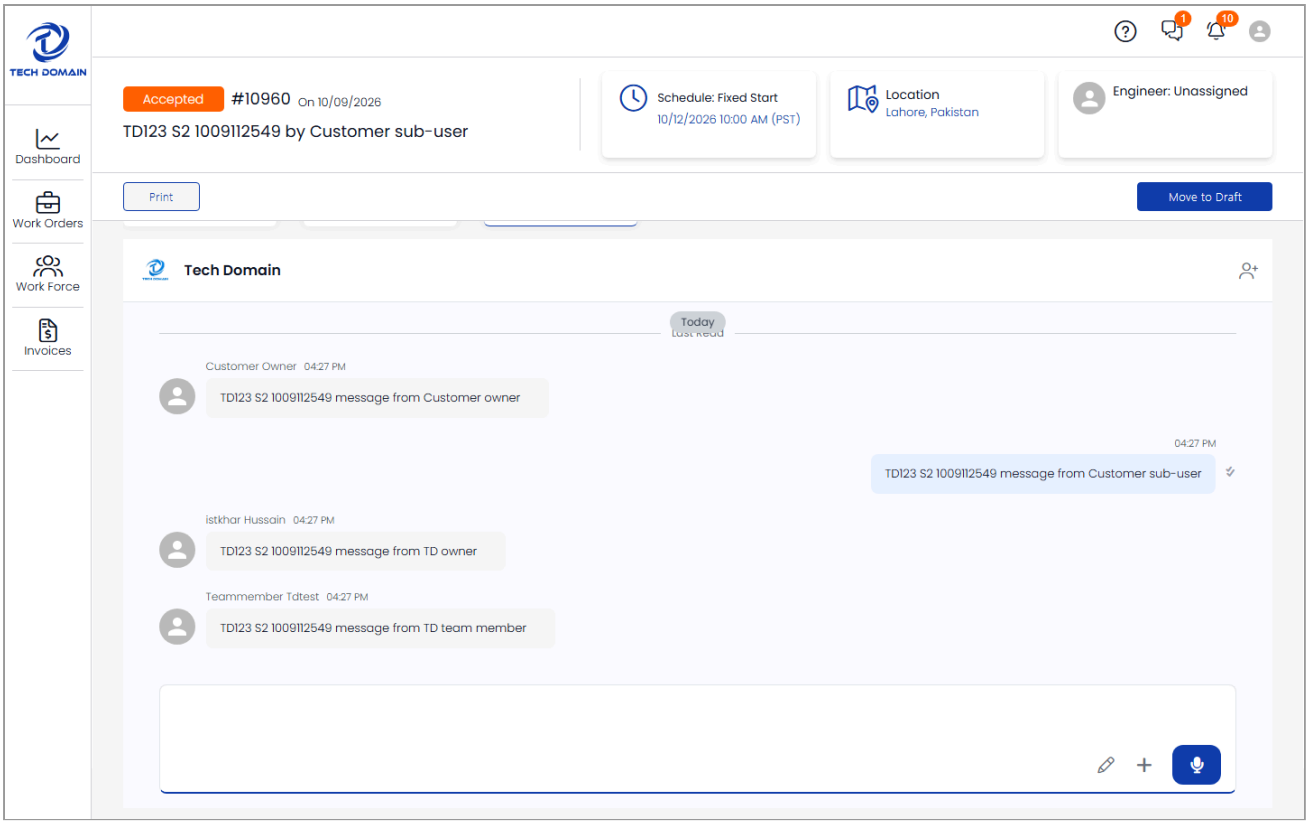
1. **Data leak - any Tech Domain company user can join a customer conversation by opening its URL.** User 641 (no team, not owner) opened /workOrder/messages/<wo>?engineerId=638 in the browser; the socket join added them as a participant. Afterwards the REST API returns the full thread to them (HTTP 200), they can post, and they receive MESSAGE notifications for every new message. Reproduced on all four work orders (961-964).
2. **Low priority / informational: the stored MESSAGE notification url is /workOrder/messages/<wo>?engineerId=null.** Manually verified on WO-10956 (2026-10-09): the web app ignores it and navigates with ?chatRoomId=<room>, so clicking the notification opens the correct customer conversation. No user impact on web; only check the mobile app if it uses the url field.
3. **Security: GET /teams/:id returns every member's bcrypt password hash** (members[].teamMember.password) to any company user.

Working as designed (not bugs): the customer sub-user is in the chat and gets notifications only for work orders the sub-user created (S2); for work orders created by the owner or by Tech Domain the sub-user gets no chat, no messages and no notifications (S1, S3, S4 - verified). Engineers and vendors near the site receive the marketplace broadcast "New work order available in Lahore..." when a work order is published. The creator never gets a notification for their own action. The engineer is never a participant of the customer room.

7. Evidence screenshots (run-1009112549)

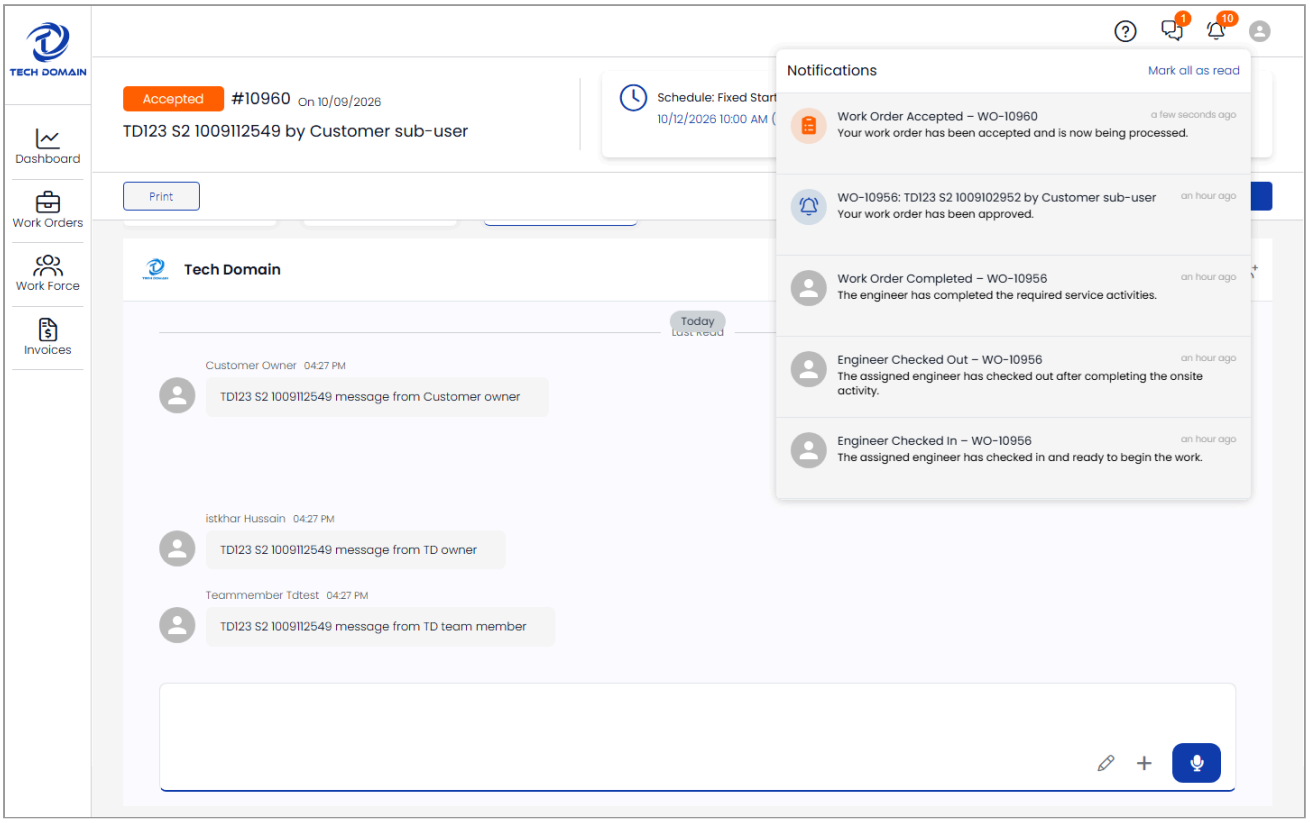
S2 - sub-user (creator) chat page: all four messages visible (owner, sub-user, TD owner, team member)

TD-123_GroupChat_14_S2-subuser-chat.png



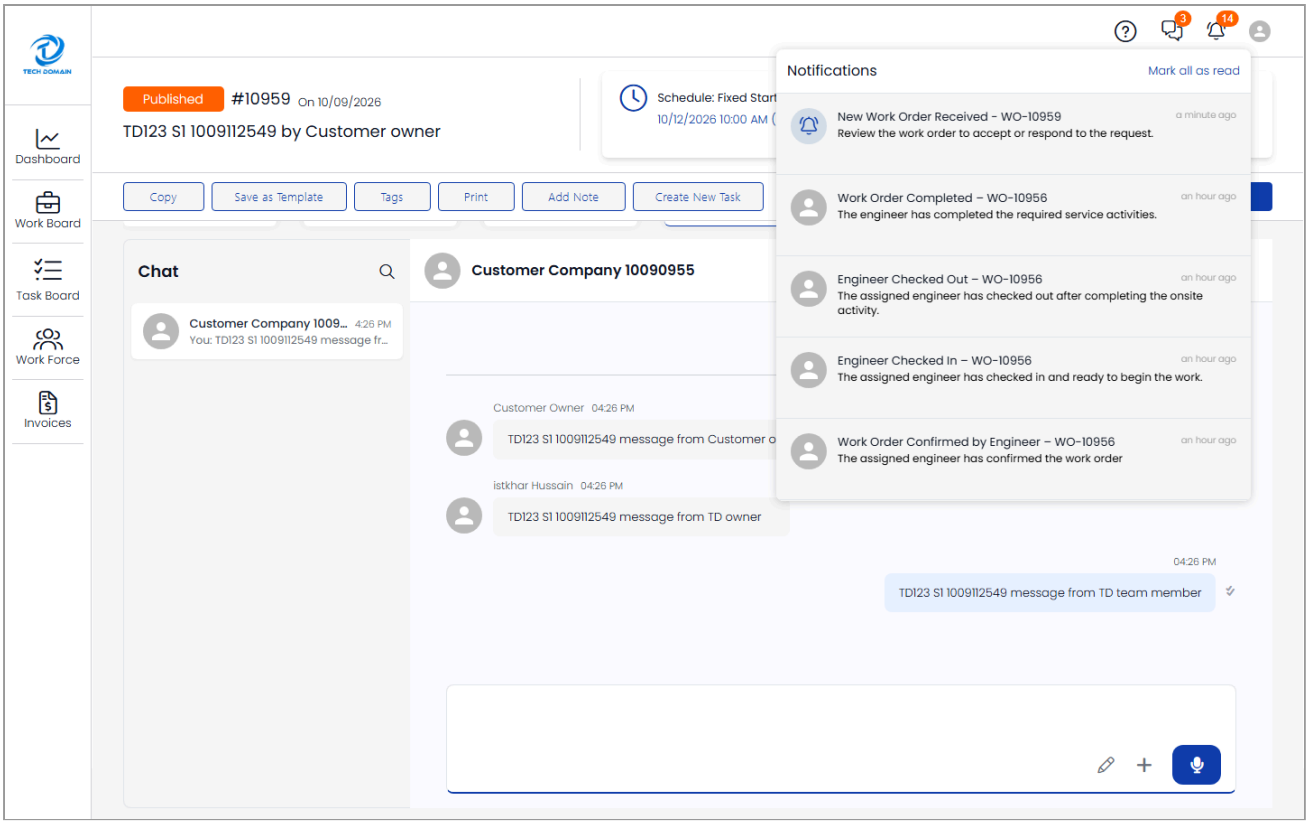
S2 - sub-user (creator) bell: "Work Order Accepted" (TD-125 pass)

TD-123_GroupChat_15_S2-subuser-bell.png



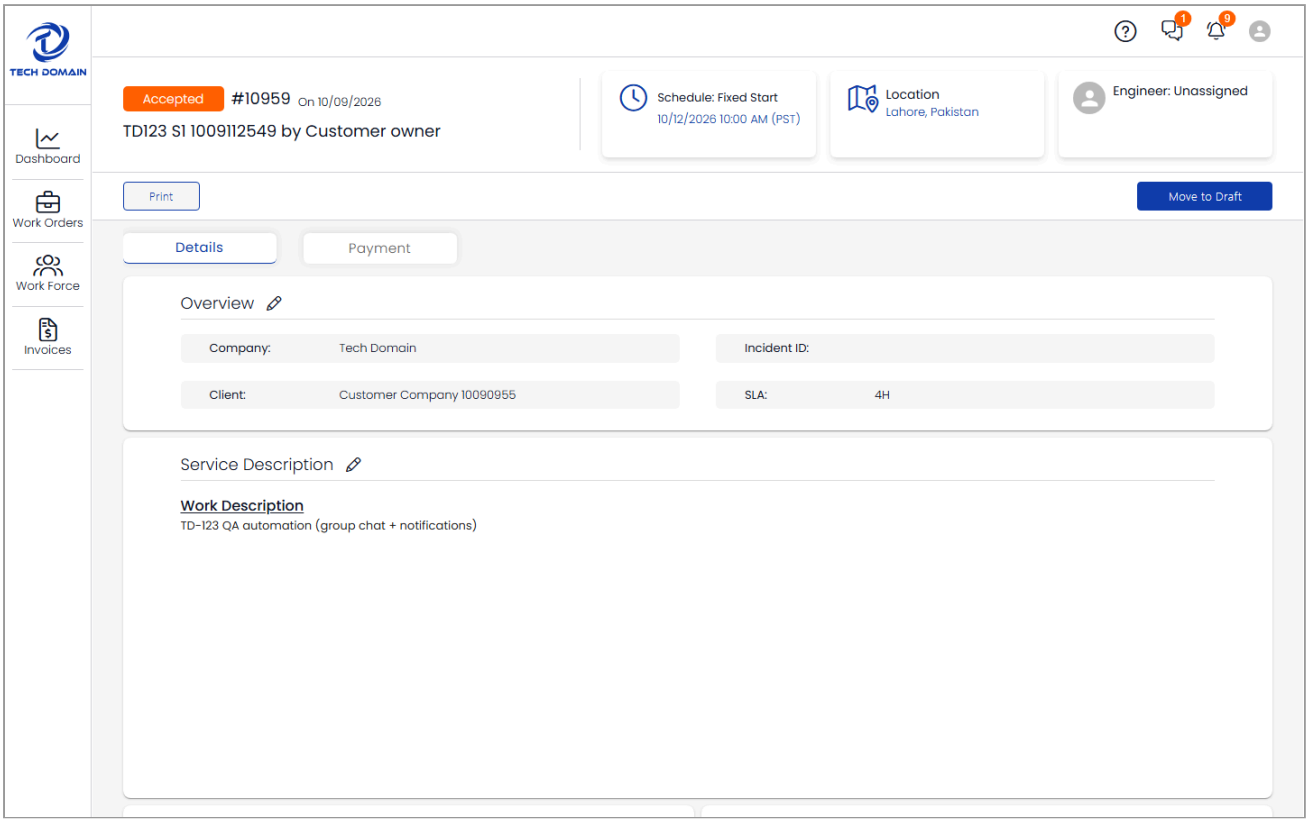
S1 - team member bell: "New Work Order Received" (TD-124 pass) and the chat with the three messages

TD-123_GroupChat_07_S1-teammember-bell.png



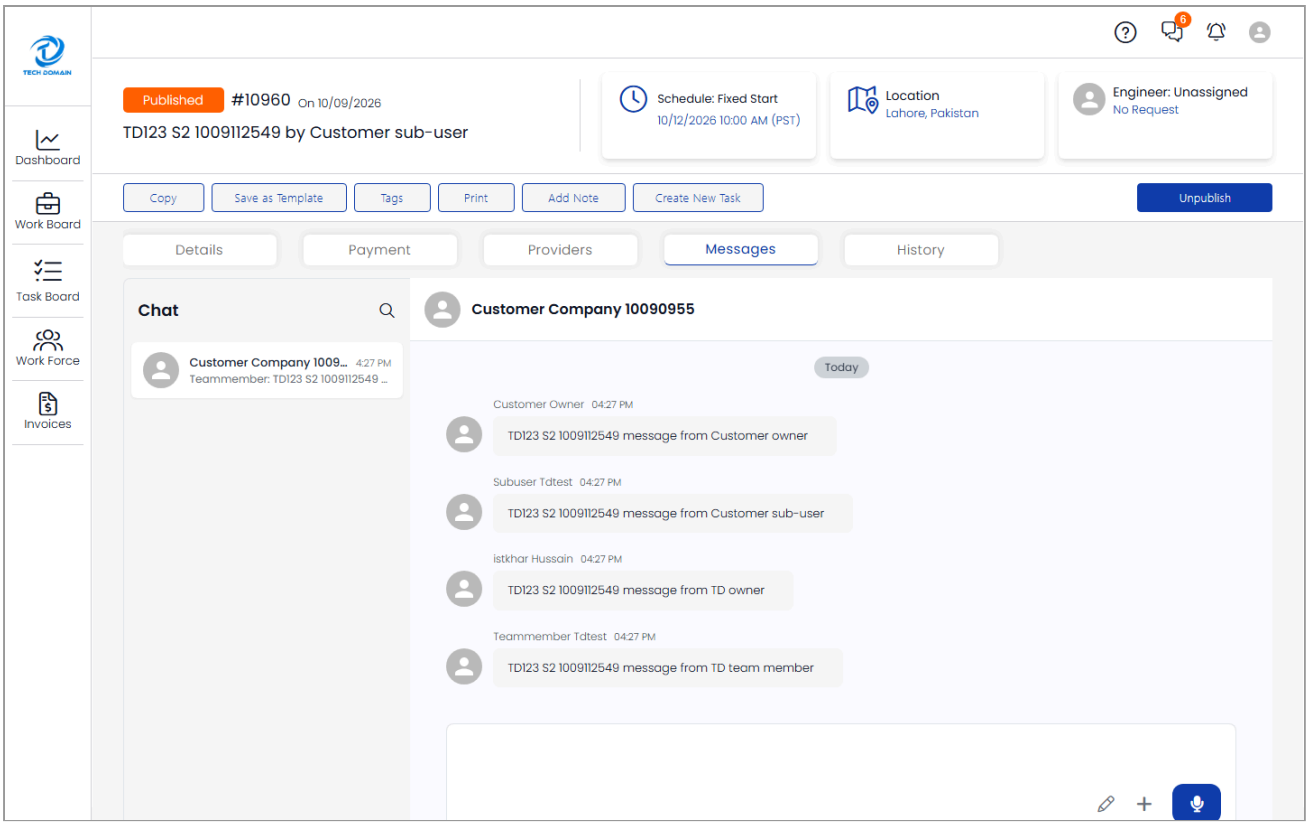
S1 - sub-user (not creator) work order page: no Messages tab / chat - expected

TD-123_GroupChat_02_S1-subuser-chat.png



S2 - LEAK: Tech Domain user outside the team opened the chat URL and sees the whole customer conversation

TD-123_GroupChat_22_S2-nonteam-chat.png



8. Dev data created or changed by this test

- Customer 103 approved by admin (branch 1, teams Service Delivery 3 + Operation 3).
- Users created: 639 (customer sub-user), 640 (TD team member, added to team Service Delivery 3), 641 (TD user without team).
- Work orders 955-964 created with "TD123 ..." titles; WO 962 walked to APPROVED with engineer 575; WO 957 assigned to engineer 575 in an earlier run.
- Evidence folder: evidence/TD-123-GroupChatNotifications/ (results.json, summary.txt and screenshots per run; REPORT.md).